

Integrated Respirometry and Metabolomics Unveil Circadian Metabolic Dynamics in *Drosophila*

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eLife Assessment

This study investigates how sleep loss and circadian disruption affect whole-organ metabolism in flies (*Drosophila melanogaster*) and reports that wild-type flies align metabolism in anticipation of diurnal rhythm, while mutant flies with impaired sleep or circadian function shift to reactive or misaligned metabolism. The integration of chamber-based flow-through respirometry with LC-MS metabolomics is innovative, and the significance of the findings is **valuable**. However, the strength of evidence needed to support the conclusions is **incomplete** based on concerns regarding the inappropriate use of constant darkness to disrupt circadian rhythms and the lack of details justifying the methods used to correlate respirometry data with whole-body metabolomics.

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Abstract

Precise temporal regulation of metabolism by sleep and circadian rhythms is essential for dynamic energy homeostasis, yet the link between systemic metabolism and respiratory demands remains poorly defined. We combined high-resolution respirometry with LC-MS-based metabolomics to characterize respiratory dynamics and metabolic states in *Drosophila melanogaster* to uncover genotype-specific impacts of sleep and circadian disruption. Wild-type flies under light-dark cycles (WT-LD) showed rhythmic respiratory patterns reflective of anticipatory coordination of mitochondrial energy metabolism, amino acid turnover, and redox cycling. In contrast, short-sleep mutants (*fmn*, *sss*) exhibited elevated metabolic rates and reactive shifts of fuel preferences toward lipid and amino acid catabolism and displayed signs of mitochondrial stress. Circadian-clock disrupted flies (*per*⁰¹, WT-DD) showed reactive and widespread metabolic dysregulation and impaired redox homeostasis. These findings demonstrate that both sleep and circadian systems are essential for aligning metabolic substrate selection with energy demands, offering mechanistic insights into how disruptions in behavioral states compromise metabolic health.

Introduction

Metabolism is a fundamental physiological process responsible for coordinating energy production and substrate utilization in response to changing physiological demands. This process is dynamically regulated by behavioral states, such as sleep and wakefulness, as well as directly by internal circadian clocks that align physiology with the external environment [1–3]. Disruptions in these temporal processes are strongly associated with metabolic disorders, including obesity and type 2 diabetes [4–6] as well as other metabolic diseases such as cardiovascular dysfunction and cancer [7].

With conserved and well-characterized sleep and circadian mechanisms [8, 9], as well as metabolic pathways also found in mammals, *Drosophila* offers a genetically tractable model for dissecting interactions between these different systems. In particular, sleep mutants with robust phenotypes can be employed to determine how loss of sleep impacts metabolic parameters, including rhythms of metabolic activity. At the same time, additional tools are now available for metabolic measurements previously not possible. Multiple approaches can be combined for a comprehensive understanding of metabolism under precisely controlled environmental conditions and at high temporal resolution [10].

Indirect calorimetry is a non-invasive form of respirometry that enables real-time quantification of whole-organism metabolic rate by measuring oxygen consumption (VO_2) and carbon dioxide production (VCO_2) [11, 12]. These measurements allow for the estimation of energy expenditure and respiratory quotient (RQ), providing insights into substrate preference (carbohydrate vs. lipid use) and metabolic flexibility [13, 14]. Although widely applied in mammalian systems [15], indirect calorimetry in small model organisms such as *Drosophila melanogaster* has only recently become feasible with advances in high-resolution, small-volume flow-through systems [16–18]. This is a critical advancement, as dissecting time-of-day-dependent metabolic regulation in mammals is often confounded by overlapping effects of sleep, activity, and feeding rhythms [19, 20]. While this can also be true in flies, the genetic tools available allow distinction between these processes to a greater extent.

This study focuses on wild-type isogenic control flies alongside three genetically characterized mutants: two short-sleep mutants: *fumin* (*fmn*), which displays impaired dopamine transporter function and defective dopamine reuptake [21] and *sleepless* (*sss*), which lacks the Sleepless protein, resulting in altered membrane excitability and reduced GABA signaling [22–24]; and the circadian mutant *period*⁰¹ (*per*⁰¹), which lacks a functional molecular clock [25]. To investigate how sleep and circadian disruption independently and interactively influence metabolic regulation, we utilized a small-animal respirometry platform built on the MAVEn-FT system, coupled with a Licor 7000 CO_2 analyzer and an Oxzilla differential O_2 analyzer. This system builds upon the Sleep and Activity Metabolic Monitor (SAMM), which first demonstrated the feasibility of simultaneous CO_2 and O_2 measurements in *Drosophila* [26]. This MAVEn-based platform enhances temporal resolution and throughput, enabling continuous measurement of respiratory parameters across multiple groups of flies over circadian time, and allowing fine-scale profiling of metabolic rhythms and their disruption across sleep-wake cycles.

To provide a more comprehensive perspective on metabolic regulation, we complemented respiratory measurements with steady-state metabolomic profiling using liquid chromatography-mass spectrometry (LC-MS) previously published from our group [27]. This integrative approach enables the identification of metabolite-level changes associated with altered energy expenditure and RQ, providing detailed insights into how sleep and circadian mutations influence metabolic homeostasis [27–29]. By employing this framework, we sought to uncover

conserved mechanisms linking temporal biological processes, such as sleep and circadian rhythms, to metabolic homeostasis and to elucidate how genetic and environmental perturbations in these systems influence energy balance and metabolic regulation.

Methods

Drosophila Strains, Entrainment and Collection

Drosophila melanogaster isogenic control (iso), two short-sleep mutants, *fumin* (*fmn*) [21] and *sleepless* (*sss*) [22], as well as a circadian clock mutant (*per*⁰¹) [25] were used for this study. Male flies were collected shortly after eclosion and entrained in light-dark (LD) incubators for a minimum of three days before circadian collection. At the time of collection, flies were between five and seven days old. Wild-type (WT) isogenic control flies were either maintained under 12 hrs:12 hrs light versus dark conditions (WT-LD) or placed in constant darkness (WT-DD) for at least 24 hours prior to collection to examine light-independent rhythms. Mutant strains (*fmn*, *sss*, and *per*⁰¹) were maintained under LD cycles. All flies were reared on a standard cornmeal/molasses medium at 25°C. Zeitgeber time (ZT) 0 was designated as lights-on, with lights-off occurring at ZT12 (12/12 LD).

Respirometry Setup

All experiments were performed using groups of 25 flies per chamber. To account for genotype-specific differences in body size, each group of 25 flies (WT, *fmn*, *sss*, and *per*⁰¹) was weighed prior to respirometry, and these weights were used to normalize VCO₂ and VO₂ values for accurate comparison. Each chamber was provisioned with 2 mL of fly food (2% agar, 5% sucrose) to sustain the flies during the 24-hour continuous recording period. Control chambers containing only food, without flies, were included to account for background microbial metabolism. To simulate natural microbial inoculation, flies were briefly introduced into these control chambers and then removed before data acquisition began.

The respirometry system continuously measured CO₂ production and O₂ consumption using a flow-through MAVEn system. It consisted of five main components: (1) a zero-grade air source and scrubbing column to remove residual CO₂, (2) mass-flow controllers to regulate airflow, (3) Sable Systems RC chambers to house the flies, (4) the MAVEn system to direct air sequentially from each chamber, and (5) gas analyzers for measuring CO₂ and O₂ concentrations. The system was calibrated every 3-5 trials using 100% N₂ (CO₂ = 0 ppm) and a known CO₂ standard to ensure measurement accuracy.

Zero-Grade Air Supply and Scrubbing

System: Zero-grade compressed air (<0.1 ppm hydrocarbons) was first passed through a custom-built scrubbing column (#26800, Drierite) to remove residual CO₂. The column was packed with an inner layer of Ascarite (#81133-20-2, Acros Organics) flanked by two outer layers of Drierite (#7779-18-9, Drierite), separated by glass wool (#11-388, Fisher Scientific).

To re-humidify the air to ~9 parts per thousand (ppt) water vapor content, it was then passed through Nafion tubing (#TT-070, Perma Pure, LLC) submerged in deionized water. Bev-A-line nonpermeable tubing (#56280, United States Plastic Corp.) was used to connect all system components.

Airflow Regulation: Re-humidified, CO₂-free air was split into two streams, each regulated by mass flow controllers to maintain a constant flow rate of 25 mL/min. One stream provided reference air to the CO₂ and O₂ analyzers (controlled by a Side-Trak 840 Series, Sierra Instruments, Inc. MFCV),

while the second stream provided flow to the fly chambers (controlled by the MAVEn's internal mass flow-controlled system).

Respirometry Chambers and MAVEn System's Airflow Management

Flies were housed in Sable Systems RC chambers (70 mm × 20 mm borosilicate glass tubes), integrated into a continuous flow through MAVEn system. The MAVEn splits airflow into 16 channels for fly chambers and one dedicated baseline channel. Airflow is constantly and continuously maintained through all fly chambers while the MAVEn's multiplexing system will switch the 'active' chamber's airflow toward the analyzer chain for a preset dwell time of 120 seconds per chamber, and 60 seconds for baseline, with a baseline interleave ratio of 4 to ensure accuracy in O₂ measurement (i.e a baseline measurement after every four chambers). In this system the MAVEn manages air flow to allow for the sequential measurements of individual chamber gas parameters and air flow rates. Experimental respirometry recording were conducted for 24-hour periods. System control and data acquisition were handled via Sable Systems MAVEn Controller software.

CO₂ Production Measurement

Baseline CO₂ levels were recorded from the reference air prior to entering the chambers. As flies respired, the CO₂ released in each chamber was subsequently flushed to the CO₂ analyzer (Li-7000, LI-COR Biosciences) via the MAVEn system.

O₂ Consumption Measurement

Similarly, O₂ consumed by flies was measured immediately using the Oxzilla differential O₂ analyzer (Oxzilla II Oxygen Analyzer, Sable Systems International) placed downstream from the CO₂ analyzer in the analyzer chain. In this setup-controlled flow reference air exiting the CO₂ analyzer's reference cell was routed into the Oxzilla's reference channel, while animal air exiting the CO₂ analyzer's sample cell was routed to the Oxzilla's sample channel. To prevent dilution effects, water vapor was removed prior to O₂ analysis using small scrubbing columns.

Water Vapor Scrubbing Columns

Separate scrubbing columns were used for reference and animal airflows and placed in-line directly prior to the Oxzilla differential O₂ analyzer's reference and sample channel inlet ports. Each column was made from a 10 mL syringe body (#14955459, Fisher Scientific) filled with an inner layer of Ascarite and two outer layers of magnesium perchlorate (M54, Fisher Scientific), separated by glass wool. Bev-A-line tubing was secured with rubber stoppers (#14-135E, Fisher Scientific). Columns were replaced twice during each 24-hour experiment to maintain performance.

Carbon Dioxide and Oxygen Analysis and Calculations

O₂ consumption was quantified by measuring the decrease in O₂ concentration as air passed through chambers containing flies. To ensure accuracy, a Savitzky-Golay filter (25-second window) was applied to smooth the raw signal, followed by a lag correction of 127 seconds accounted for delay between chamber exit and O₂ analyzer input. To match dry atmospheric air, the reference air was baseline corrected to 20.95% O₂. For both reference and animal air the %O₂ values were converted to O₂ fractional contents by dividing percentage values by 100 (thus baseline air has a fractional content of 0.2095). The VO₂ value from an empty chamber was subtracted from experimental values [30]:

$$VO_2 = (\text{Animal } O_2 - \text{Reference } O_2) \times \text{Flow rate} / (1 - \text{Reference } O_2)$$

VO₂ values were expressed in μL/hr by summing all 5-minute bins into hourly averages.

CO₂ production was quantified by measuring the increase in CO₂ concentration as air passed through chambers containing flies. To ensure accuracy, a Savitzky-Golay filter (25-second window) was applied to smooth the raw signal, followed by a lag correction of 22 seconds with Z correction and 28 seconds without Z correction accounted for delay between chamber exit and O₂ analyzer input. To match dry atmospheric air, the reference air was baseline corrected to 0 parts per million (ppm) CO₂. For both reference and animal air the ppm CO₂ values were converted to CO₂ fractional contents by dividing percentage values by 1000000. The VCO₂ value from an empty chamber was subtracted from experimental values [30]:

$$VCO_2 = (\text{Reference CO}_2 + \text{Animal CO}_2) \times \text{Flow rate} / 1 - \text{Reference CO}_2$$

VCO₂ values were expressed in μL/hr by summing all 5-minute bins into hourly averages.

Respiratory quotient (RQ) was calculated as:

$RQ = VCO_2 / VO_2$. This provided a ratio of CO₂ produced to O₂ consumed, reflecting macronutrient oxidation and metabolic flexibility.

Metabolite Extraction and LCMS Measurements

The metabolomics dataset analyzed in this study was previously published and is publicly available, as described in detail in [27, 31]. Briefly, polar metabolites were extracted from fly bodies sampled every 2 hours from ZT0 to ZT22 using a modified Bligh-Dyer extraction method, as previously reported [31, 32]. The polar fraction of each extract was dried under vacuum and reconstituted in 100 μL of acetonitrile:Milli-Q water, followed by vortexing for 20 seconds.

Samples were analyzed using an ion-switching LC-MS method, and peak integration and data processing were performed according to previously published procedures [27].

Statistical Analysis

All statistical analyses were conducted using GraphPad Prism. Non-parametric Spearman correlation was used to evaluate relationships among metabolic parameters. Metabolites with *p*-values below 0.05 were considered statistically significant and selected for further investigation. To assess genotype-specific differences, one-way ANOVA was performed followed by a pairwise unpaired two-tailed Student's *t*-tests comparing the control group (WT-LD) with each genotype (*fmn*, *sss*, *per*⁰¹, and WT-DD).

Rhythmicity analysis

Rhythmicity analysis was performed using Nitecap, a tool for circadian and rhythmic analyses [33]. Time-series metabolic data were analyzed to compute rhythmic parameters using the RAIN algorithm [34]. Rhythms were assessed for statistical significance using false discovery rate (FDR)-adjusted *p*-values, ensuring robust detection while minimizing false positives. The lag parameters of each rhythm were computed using JTK cycle algorithm [35].

Pathway Analysis

Significant metabolites identified from univariate analyses were used for metabolic pathway analysis via MetaboAnalyst 5.0. Metabolites were uploaded using HMDB identifiers and analyzed using the hypergeometric enrichment method, with relative-betweenness centrality applied for topology analysis based on the *Drosophila melanogaster* (KEGG) pathway library. Pathways with a *p*-value less than 0.05 were considered statistically significant.

Results

Metabolic Fuel Utilization Across Genotypes Based on Respiratory Quotient (RQ)

To assess whole-body metabolic activity, we performed respirometry on groups of 25 flies per genotype, measuring oxygen consumption (VO_2) and carbon dioxide production (VCO_2) across the circadian cycle. The respiratory quotient (RQ), calculated as the ratio of VCO_2 to VO_2 , provides insight into the predominant metabolic substrate being utilized: an RQ of 1.0 reflects pure carbohydrate metabolism, values below 1.0 indicate lipid and protein oxidation, and values above 1.0 suggest lipogenesis (the conversion of carbohydrates into fats) [14]. RQ, VCO_2 , and VO_2 were continuously recorded from Zeitgeber Time (ZT) 0 to 24 at one-second intervals and subsequently averaged into 5-minute bins for analysis. **Figures 1a-c** shows the 5-minute binned VCO_2 , VO_2 and RQ profiles across the circadian cycle for all genotypes (separate individual plots for VCO_2 , VO_2 , and RQ for each genotype are provided in the Figure S1a-e, S2a-e and S3a-e respectively). The average values of VCO_2 , VO_2 and RQ across the full 24-hour cycle are summarized in **Figure 1d-f**.

Among the genotypes tested, WT-LD, WT-DD, and *per*⁰¹ flies exhibited similarly elevated RQ values (1.20, 1.19, and 1.20, respectively), consistent with active lipogenesis. Group differences were assessed using the Kruskal–Wallis test, followed by Dunn’s multiple comparisons post hoc test revealed no statistically significant differences in RQ among these three genotypes (**Figure 1f**, *ns*), suggesting that circadian disruption in constant darkness or in the *per*⁰¹ mutant does not alter the dominant fuel utilization strategy under basal conditions.

In contrast, short-sleeping mutants displayed lower RQ values. The *fmn* mutant exhibited a moderate reduction (RQ = 1.09), reflecting increased reliance on carbohydrate metabolism. The *sss* mutant showed the lowest RQ (0.94), indicating a shift toward lipid and protein catabolism. Both mutants were significantly different from WT-LD ($p < 0.001$), highlighting genotype-specific alterations in fuel utilization associated with sleep loss (**Figure 1f**).

These results demonstrate that while WT and circadian-disrupted genotypes maintain a lipogenic profile, sleep-deficient mutants exhibit altered substrate utilization, favoring catabolism of carbohydrates or lipids depending on the severity of sleep disruption.

Furthermore, to account for potential differences in body size that could confound VCO_2 and VO_2 measurements, fly weights were recorded for each genotype before respirometry. Notably, *fmn* and *sss* mutants weighed significantly less than WT controls, whereas *per*⁰¹ flies had comparable weights (**Figure 1g**). Accordingly, VCO_2 and VO_2 values were normalized to fly weight, and the genotype-specific differences remained robust after normalization (**Figure 1h-i**). These data confirm that the observed metabolic differences are not attributable to body size but reflect inherent alterations in metabolic fuel utilization.

Diurnal Variation in CO_2 Production, O_2 Consumption, and Respiratory Quotient Across Genotypes

To investigate how sleep and circadian regulation influence temporal patterns of metabolism, we measured carbon dioxide production (VCO_2), oxygen consumption (VO_2), and respiratory quotient (RQ) across day and night phases in wild-type flies maintained under light-dark cycles (WT-LD), wild-type flies kept in constant darkness (WT-DD), short-sleep mutants (*fumin* [*fmn*] and sleepless [*sss*]), and the circadian clock mutant *period*⁰¹ (*per*⁰¹). The average values of VCO_2 , VO_2 and RQ during the day (ZT0-12) and night (ZT12-24) are presented in Figure S4.

VCO_2 , VO_2 and Respiratory quotient (RQ) profiles across the circadian cycle in different genotypes

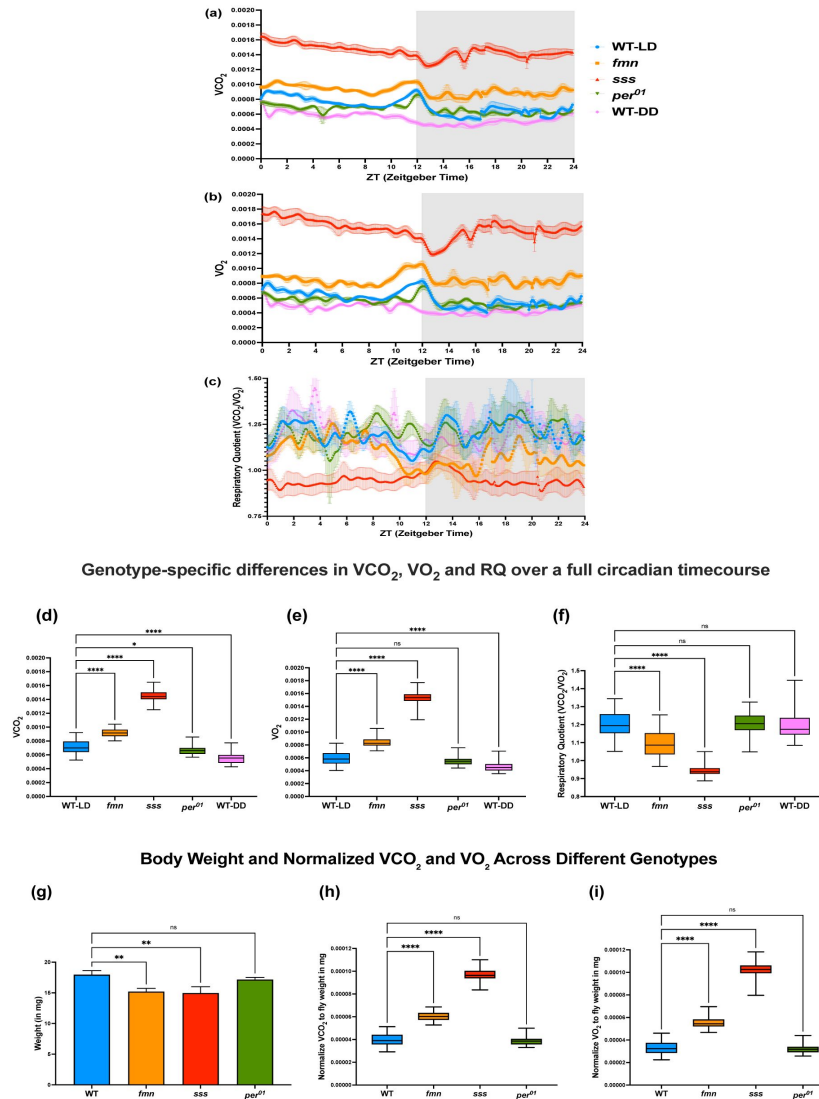


Figure 1.

a-c. VCO_2 , VO_2 and Respiratory quotient (RQ) profiles across the circadian cycle in different genotypes. VCO_2 , VO_2 and RQ (VCO_2/VO_2) was continuously recorded at one-second intervals from Zeitgeber Time (ZT) 0 to 24 and subsequently averaged into 5-minute bins for analysis. Traces represent mean VCO_2 , VO_2 and RQ values across the circadian cycle for wild-type flies under light-dark conditions (WT-LD), short-sleep mutants *fumin* (*fmn*) and *sleepless* (*sss*), circadian clock mutant *period⁰¹* (*per⁰¹*), and wild-type flies maintained in constant darkness (WT-DD). **d-f. Genotype-specific differences in VCO_2 , VO_2 and RQ measured over a full circadian timecourse.** Boxplots show average values of (left to right) respiratory quotient (RQ), carbon dioxide production (VCO_2), and oxygen consumption (VO_2) across genotypes and lighting conditions. Measurements were taken continuously over a 24-hour period using a flow-through MAVEn system. Genotypes include wild-type under light-dark (WT-LD) and constant darkness (WT-DD), short-sleep mutants *fumin* (*fmn*) and *sleepless* (*sss*), and circadian mutant *per⁰¹*. Group differences were assessed using the **Kruskal-Wallis test**, followed by **Dunn's multiple comparisons post hoc test**. Significance is denoted as: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***); ns = not significant. **g-i. Body Weight and Normalized VCO_2 and VO_2 Across Different Genotypes.** Body weight (mg) and respiratory parameters (VCO_2 and VO_2) normalized to body weight (mg) were measured in wild type (WT), *fmn*, *sss*, and *per⁰¹* mutant flies. Statistical significance was assessed using one-way ANOVA followed by Dunn's multiple comparisons test against WT. $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***); ns = not significant.

The *sss* mutant exhibited the highest metabolic rates among all genotypes, with significantly elevated VCO_2 and VO_2 during both day and night relative to WT-LD ($p < 0.001$). A pronounced day-night difference in *sss* further indicated a strong phase-dependent increase in respiratory activity (Figure S4). The *fumin* mutant also showed significantly elevated VCO_2 and VO_2 compared to WT-LD ($p < 0.001$), with a detectable but less marked diurnal variation than in *sss* (Figure S4). WT-LD flies exhibited a robust diurnal rhythm, with significantly higher respiratory rates during the day ($p < 0.001$), consistent with normal circadian regulation of energy expenditure (Figure S4).

The circadian mutant *per*⁰¹, lacking a functional molecular clock, displayed an attenuated and phase-altered respiratory pattern. Both VCO_2 and VO_2 were significantly reduced during the day ($p < 0.05$), with no significant changes at night, indicative of disrupted or misaligned metabolic rhythmicity (Figure S4). WT-DD flies, maintained in constant darkness, exhibited the lowest overall respiratory activity. Despite the absence of environmental light cues, VCO_2 and VO_2 showed significant day-night differences, while RQ remained unchanged during the day but displayed a modest difference at night ($p < 0.05$), indicating limited circadian control over metabolic rhythms in constant conditions (Figure S4).

Circadian Variations in CO₂ Production, O₂ Consumption, and Respiratory Quotient Across Genotypes

To evaluate genotype-dependent circadian regulation of metabolism, we assessed rhythmicity in carbon dioxide production (VCO_2), oxygen consumption (VO_2), and respiratory quotient (RQ) in wild-type flies under light-dark conditions (WT-LD), wild-type flies in constant darkness (WT-DD), short-sleep mutants (*fumin* [*fmn*] and *sleepless* [*sss*]), and the circadian clock mutant *period*⁰¹ (*per*⁰¹). Time-course data were analyzed using JTK CYCLE and RAIN algorithms, with rhythmicity classified as statistically significant ($p \leq 0.05$), trending ($0.05 < p < 0.1$), or not significant ($p \geq 0.1$). Circadian phase was estimated using JTK lag values (Figures 2–3, Table 1).

VCO_2 rhythms were statistically significant in WT-LD, WT-DD, *fmn*, *sss* and *per*⁰¹ (Figure 2, Table 1). These findings demonstrate that VCO_2 exhibits robust circadian oscillations under both light-dark and constant darkness conditions, and that rhythmicity persists even in the absence of a functional *period* gene. VCO_2 peak occurred near ZT ~4 in WT-LD, ZT ~7.5 in WT-DD, ZT ~3.25 in *fmn*, ZT ~3 in *per*⁰¹, and ZT ~2 in *sss*, indicating genotype-specific shifts in respiratory phase (Figure 3, Table 1).

VO_2 rhythmicity was statistically significant in WT-LD, WT-DD, *sss* and *per*⁰¹ while not significant in *fmn* (Figure 2, Table 1). These findings indicate that circadian regulation of oxygen consumption is evident in constant darkness and in the absence of a functional *period* gene but is reduced in sleep mutant. VO_2 peaked near ZT ~4 in WT-LD, ZT ~7 in WT-DD, ZT ~1.25 in *sss* and ZT ~6 in *per*⁰¹, suggesting conserved phase alignment (Figure 3, Table 1).

RQ also showed genotype-dependent rhythmicity, with significant oscillations observed in *fmn*, *sss* and *per*⁰¹, while no significant rhythms were observed in WT-LD or WT-DD (Figure 2, Table 1). In the rhythmic genotypes, RQ peaked at ZT ~4.25 in *fmn*, ZT ~2 in *per*⁰¹ and ZT ~12.5 in *sss* indicating distinct phase alignment compared to respiratory output (Figure 3, Table 1).

Temporal Profiling of Respiratory Quotient in Wild-Type Flies Under Light-Dark Conditions

To align respiratory quotient (RQ), which was collected at one-second intervals and averaged into 5-minute bins, with steady-state metabolite measurements taken at 2-hour intervals, RQ values corresponding to each 2-hour Zeitgeber Time (ZT) point were extracted from the 5-minute binned dataset. Building on this alignment, we next explored temporal relationships at higher resolution by performing a 2-hour lag analysis, systematically shifting the RQ time series by -120, -60, -30,

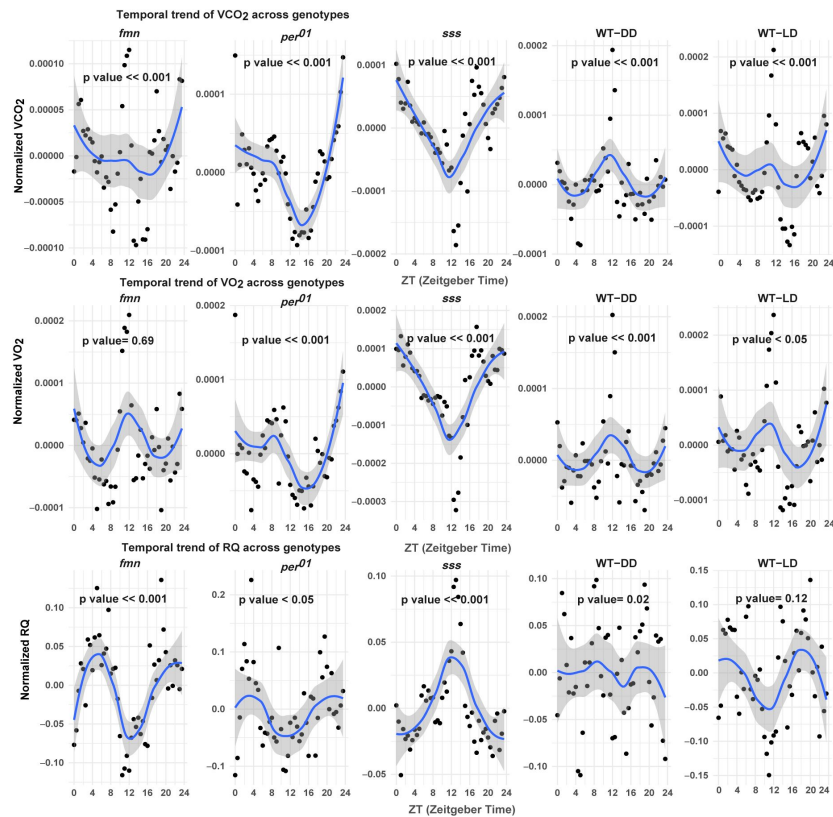


Figure 2.

Circadian timecourse of respiratory parameters across genotypes.

Normalized temporal profiles of carbon dioxide production (VCO_2) (top), oxygen consumption (VO_2) (middle), and respiratory quotient (RQ) (bottom) plotted over a 24-hour circadian cycle (ZT0–24) for genotypes: wild-type in light-dark (WT-LD) and constant darkness (WT-DD), short-sleep mutants *fumin* (*fmn*) and *sleepless* (*sss*), and circadian mutant *per⁰¹*. Curves reveal genotype-specific rhythmicity and metabolic dynamics across circadian time. Significant genotype $\times$ time interactions were observed for all variables (VCO_2 , VO_2 , RQ), indicating altered circadian rhythmicity in mutants. $p < 0.05$ considered significant.

Figure 3.

Circadian phase distribution of respiratory rhythms across genotypes.

Polar plots depicting the phase (peak timing) of rhythmic expression for carbon dioxide production (VCO_2), oxygen consumption (VO_2), and respiratory quotient (RQ) over a 24-hour cycle (ZT0–24) for each genotype. Genotypes include wild-type in light-dark (WT-LD) and constant darkness (WT-DD), short-sleep mutants *fumin* (*fmn*) and *sleepless* (*sss*), and circadian mutant *per⁰¹*. Statistical significance of rhythmicity was determined using the RAIN algorithm: darker-colored bars indicate significant rhythms ($p < 0.05$), and lighter-colored bars indicate non-significant rhythms ($p > 0.05$).

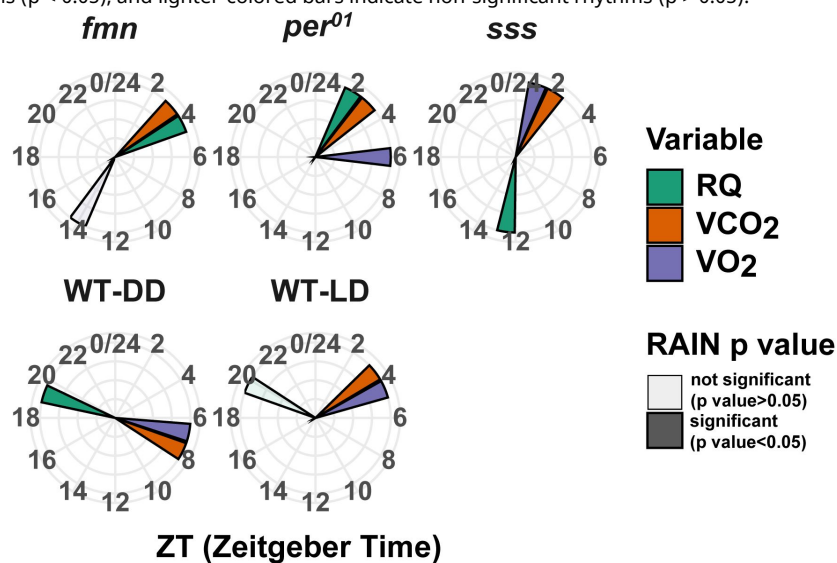


Table 1.

Circadian rhythmicity metrics for respiratory parameters across genotypes

Genotype	RQ JTK Lag	RQ JTK Period	RQ RAIN p-value	VCO_2 JTK Lag	VCO_2 JTK Period	VCO_2 RAIN p-value	VO_2 JTK Lag	VO_2 JTK Period	VO_2 RAIN p-value
WT-LD	19.75	20	0.12	4	24	1.1143e-05	4	24	0.001
<i>fumin</i>	4.25	22.5	9.54e-08	3.25	24	0.0002	14	20	0.69
<i>sss</i>	12.5	20	1.41e-07	2	24	1.33e-09	1.25	24	1.14e-10
<i>per⁰¹</i>	2	20	0.006	3	24	1.05e-13	6	21.5	1.82e-07
WT-DD	19.25	23	0.02	7.5	24	0.0002	7	24	9.4145e-06

-15, -5, +5, +15, +30, +60, and +120-minutes relative to the metabolite dataset. This analysis revealed a distinct circadian rhythmicity in RQ, characterized by oscillatory patterns indicative of coordinated substrate utilization across the day-night cycle (**Figure 4** [↗](#)).

Correlation of Respiratory Quotient with Metabolite Profiles and Temporal Lag Analysis in WT- LD Flies

To investigate the association between respiratory quotient (RQ) and metabolite in WT-LD flies, we initially performed Spearman correlation analysis (ρ) across a range of temporal lags (-120 to +120 minutes). Several metabolites demonstrated strong correlations with RQ ($|\rho| > 0.7$, $p < 0.05$), suggesting tight coupling between metabolic fluctuations and respiratory output. To minimize the influence of outliers and enhance data visualization, correlation patterns were subsequently presented as rank-based scatter plots. Representative examples are shown in **Figure 5a** [↗](#): Hydroxyhexadecenoylcarnitine exhibited a significant positive correlation with RQ ($\rho = +0.78$, $p < 0.05$) at a +120-minute lag, while Quinolate showed a strong negative correlation ($\rho = -0.77$, $p < 0.05$) at a -120-minute lag.

To further characterize these associations, metabolites were categorized based on the timing of their peak changes relative to RQ fluctuations, exhibiting either positive or negative lag. To enable direct comparison of temporal dynamics between RQ and metabolite profiles across ZT 0 to ZT 24, z-scoring was performed to normalize differences in absolute magnitude and measurement units. Within this framework, a positive lag indicates that metabolite changes occur after shifts in RQ, requiring the metabolite profile to be shifted forward (rightward) along the time axis for optimal alignment. Conversely, a negative lag signifies that metabolite changes precede RQ fluctuations, necessitating a backward (leftward) shift of the metabolite profile to achieve alignment (**Figure 5b** [↗](#)).

Metabolite Response Dynamics Relative to RQ and Functional Pathway Enrichment of RQ- Associated Metabolites in WT-LD Flies

To further explore the temporal relationship between the metabolites and respiratory quotient (RQ), we generated a clustered heatmap of metabolites significantly correlated with RQ. Metabolites were filtered based on statistical significance ($|\rho| > 0.7$, $p < 0.05$ at any timepoint), and their temporal profiles were realigned relative to respiration, anchoring RQ at $t = 0$ (**Figure 6** [↗](#)). This flipping step was essential to standardize the visualization of metabolites either preceding or following respiratory changes, facilitating clearer interpretation of biological timing relationships.

To identify key metabolic pathways associated with circadian regulation of respiration, we next performed pathway enrichment analysis on the RQ-correlated metabolites using MetaboAnalyst. Enriched pathways were identified at a significance threshold of $p < 0.05$. WT-LD-specific pathway enrichment results are presented in **Figure 6** [↗](#). Consistent with these findings, WT flies under LD conditions exhibited circadian-regulated metabolic rhythms, characterized by coordinated activity in amino acid metabolism, redox balance, and energy-generating pathways such as the TCA cycle and glyoxylate metabolism.

RQ-Linked Metabolic Shifts and Pathway Enrichment in Sleep Mutants

We applied the same analysis method to sleep mutant flies (*fmn* and *sss*) to identify metabolic pathways associated with altered respiratory rhythms. Metabolites significantly correlated with RQ ($|\rho| > 0.7$, $p < 0.05$) were identified, and their temporal profiles were realigned relative to respiration by anchoring RQ at $t = 0$. Clustered heatmaps were generated to visualize metabolite response dynamics, and pathway enrichment analysis of RQ-associated metabolites was

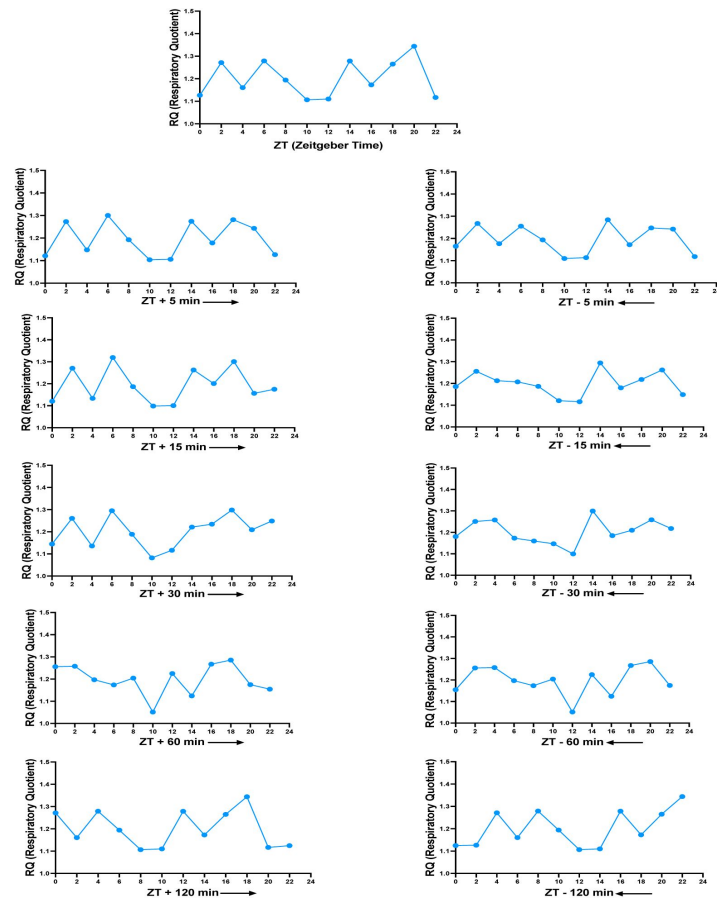
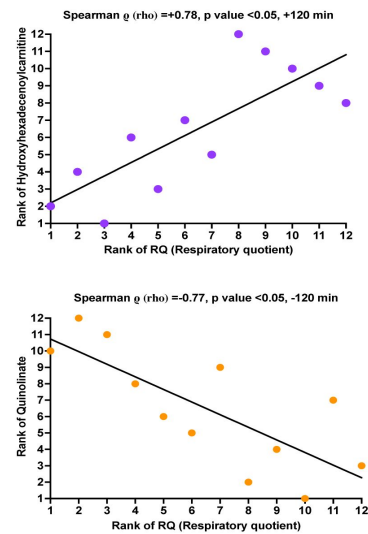


Figure 4.

Temporal Profiling of Respiratory Quotient in Wild-Type Flies Under Light-Dark Conditions.

Respiratory quotient (RQ) was extracted every 2 h across a 24-hour light-dark (LD) cycle in WT flies. Lag analyses were performed by advancing or delaying the RQ data by -120, -60, -30, -15, -5, +5, +15, +30, +60-, and +120-minutes relative to Zeitgeber Time (ZT). Each panel represents the RQ profile corresponding to a specific time shift, illustrating the phase-dependent dynamics of respiration across the circadian cycle.

a. Rank-based correlation between Respiratory Quotient (RQ) and selected metabolites



b. Temporal alignment of metabolite with respiratory quotient (RQ) in WT-LD flies

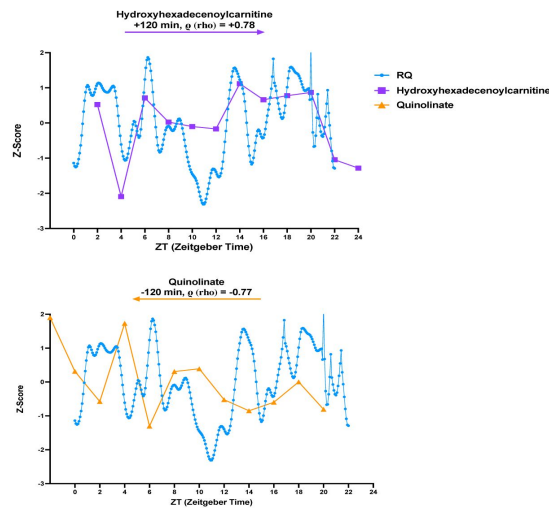


Figure 5.

a. Rank-based correlation between Respiratory Quotient (RQ) and selected metabolites. Scatter plots depict the rank-order relationships between RQ and (A) Hydroxyhexadecenoylcarnitine and (B) Quinolate. Each point represents a timepoint after aligning data with respective time shifts. Spearman's rank correlation coefficient (ρ) and corresponding p-values are indicated on each panel. A positive lag (+120 minutes for Hydroxyhexadecenoylcarnitine) or negative lag (-120 minutes for Quinolate) denotes the temporal shift in RQ relative to the metabolite dataset. Statistical significance was defined as $p < 0.05$. **b. Temporal alignment of metabolite with respiratory quotient (RQ) in WT-LD flies.** Z-scored time series of RQ, Hydroxyhexadecenoylcarnitine, and Quinolate across a 24-hour light-dark cycle. **Top Panel:** Hydroxyhexadecenoylcarnitine shows a positive lag of +120 minutes and strong positive correlation with RQ ($\rho = +0.78$), suggesting its rise after changes in RQ. **Bottom Panel:** Quinolate shows a negative lag of -120 minutes and strong negative correlation with RQ ($\rho = -0.77$), indicating it precedes changes in RQ.

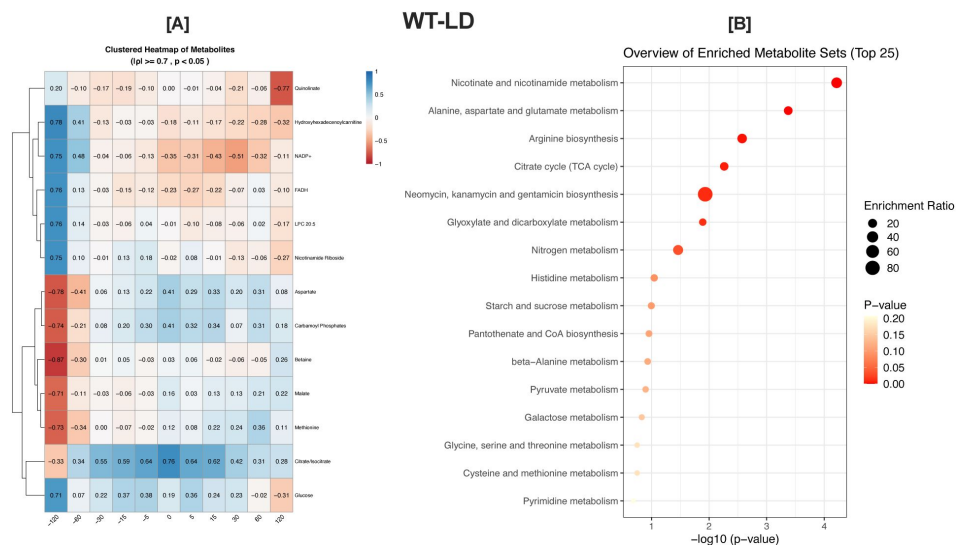


Figure 6.

Metabolite-Respiratory Quotient Correlations and Pathway Enrichment in WT-LD Flies.

(A) Heatmap depicting Spearman correlations ($|p| > 0.7, p < 0.05$) between respiratory quotient (RQ) and metabolite across the circadian cycle in wild-type flies maintained under light-dark (LD) conditions. Metabolites were clustered based on correlation patterns, highlighting groups with similar temporal associations with respiratory activity. (B) Pathway enrichment analysis of significantly correlated metabolites, illustrating metabolic pathways most closely linked to respiratory dynamics. Pathways with a p-value less than 0.05 were considered statistically significant.

conducted using MetaboAnalyst ($p < 0.05$) (Figure 7). This analysis revealed that sleep mutants exhibit genotype-specific metabolic disruptions implicating mitochondrial and energy-related pathways. In *fmn* flies, alterations were observed in arginine biosynthesis, purine and nitrogen metabolism, and butanoate pathways, suggestive of dysregulated nitrogen balance and mitochondrial dysfunction. In contrast, *sss* mutants showed perturbations in glycine, serine, and threonine metabolism, as well as carbohydrate and antibiotic biosynthetic pathways, reflecting altered carbon utilization and mitochondrial-associated metabolic stress.

RQ-Linked Metabolic Shifts in Circadian Mutants and WT Flies in Constant Darkness

Circadian clock mutant flies (*per*⁰¹) and wild-type flies maintained under constant darkness (WT-DD) were analyzed using the same approach. Metabolites exhibiting strong correlations with RQ ($|\rho| > 0.7$, $p < 0.05$) were identified and temporally aligned by anchoring RQ at $t = 0$. Clustered heatmaps were constructed to visualize the dynamic responses of these metabolites, and pathway enrichment analysis of RQ-associated metabolites was performed using MetaboAnalyst ($p < 0.05$) (Figure 8). Both *per*⁰¹ and WT-DD flies exhibited disrupted temporal coordination of mitochondrial metabolism. *per*⁰¹ mutants showed dysregulation in amino acid metabolism, purine turnover, and redox-associated pathways such as glutathione and nicotinate metabolism, consistent with impaired redox buffering and energy imbalance. In contrast, WT-DD flies exhibited alterations in sulfur- and nitrogen-containing amino acid pathways, indicative of desynchronized but intact metabolic cycling in the absence of external light cues.

Discussion

Temporal Misalignment Alters Fuel Utilization and Respiratory Rhythms

Our integrative approach, combining whole-organism respirometry [18, 26] with LC-MS-based metabolomics [27, 36–40], reveals how sleep and circadian disruptions distinctly alter energy metabolism in *Drosophila melanogaster*. By comparing wild-type flies with mutants affecting sleep (*fumin* and *sleepless*) [21, 22, 41, 42] and circadian clock function (*per*⁰¹) [24], we have captured genotype-specific differences in respiratory dynamics and substrate usage across the circadian cycle under tightly controlled environmental conditions. These findings establish a functional framework for understanding how behavioral state and clock integrity shape metabolic homeostasis.

Our data suggest that sleep deprivation and circadian disruption drive distinct changes in whole-organism respiratory physiology and fuel selection. Metabolites in wild-type LD typically showed peak correlations preceding respiratory changes (negative lag), reflecting anticipatory circadian control. In contrast, mutants displayed a shift to reactive correlations (positive or zero lag), indicating loss of temporal synchronization and proactive metabolic adjustments. These phenotypes closely parallel metabolic outcomes reported in mammalian systems, where disruptions in behavioral states such as sleep deprivation or circadian misalignment are linked to increased energy expenditure, altered substrate utilization patterns, and systemic metabolic imbalance [3, 43]. In both rodents and humans, sleep loss leads to elevated basal metabolic rate and a shift toward carbohydrate oxidation and lipid/protein catabolism [3, 43, 44]. A similar shift is evident in our study, where *Drosophila* short-sleep mutants-*fmn* flies display moderately reduced RQ values (1.09) and *sss* mutants show a more pronounced shift (0.94)-exhibit enhanced catabolic metabolism.

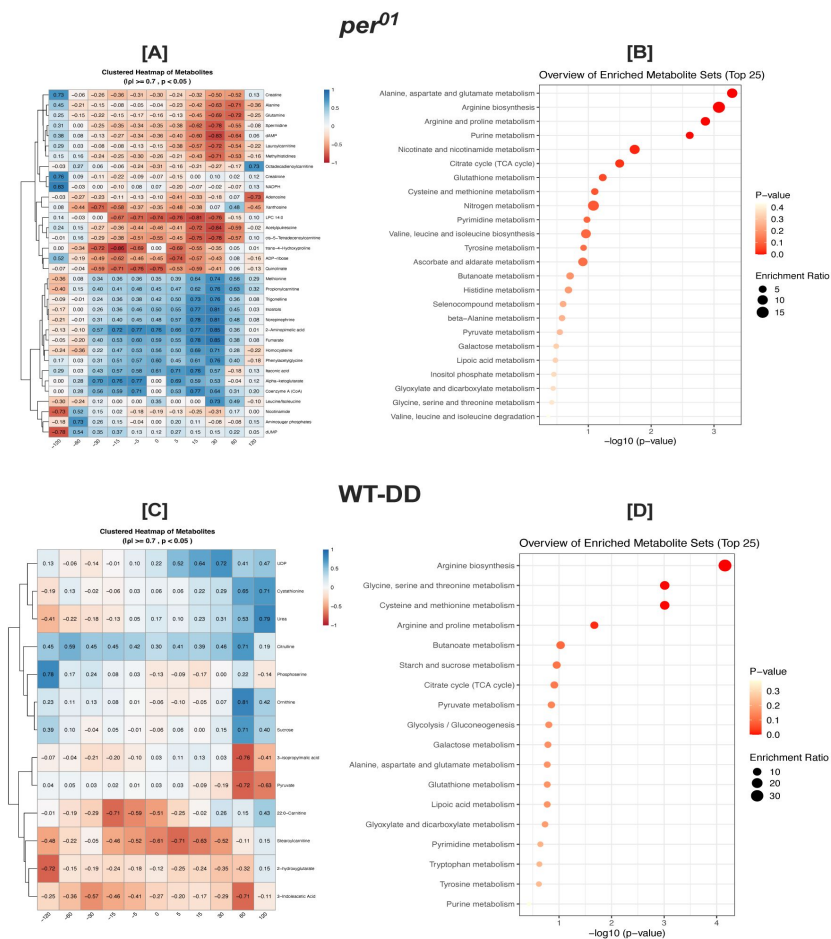


Figure 8.

Correlation of Metabolites with Respiratory Quotient and Pathway Enrichment in Circadian Mutant and Wild-Type Flies in Constant Darkness.

(A, C) Heatmaps displaying Spearman correlations ($|p| > 0.7, p < 0.05$) between respiratory quotient (RQ) and metabolites in the circadian mutant *per⁰¹* (A) and wild-type flies maintained in constant darkness (WT-DD) (C). (B, D) Pathway enrichment analyses of metabolites significantly correlated with RQ in *per⁰¹* (B) and WT-DD (D). Pathways with p-values less than 0.05 were considered statistically significant.

Likewise, the dampened and phase-shifted respiratory rhythms observed in *per*⁰¹ and WT-DD flies resemble patterns seen in mammalian models of circadian misalignment, which exhibit blunted respiratory oscillations, disrupted glucose and lipid regulation, and increased risk for metabolic disease [45 [↗](#), 46 [↗](#)]. These conserved physiological responses underscore the translational value of *Drosophila* as a model system for dissecting the interplay between sleep, circadian timing, and metabolic regulation.

Wild-Type Flies Exhibit Circadian-Regulated Biosynthesis and Redox Balance Under LD Cycles

In wild-type flies maintained under a 12:12 light-dark (LD) cycle, RQ-correlated metabolites revealed tightly coordinated circadian regulation of biosynthetic and redox pathways. An increase in nicotinate and nicotinamide metabolism-highlighted by changes in quinolinate, nicotinamide riboside, and L-aspartic acid-suggests a direct link between mitochondrial respiration and NAD⁺ biosynthesis [47 [↗](#), 48 [↗](#)]. These NAD⁺ precursor changes occur before respiration increases (negative lag), highlighting anticipatory NAD⁺ regulation. Concurrent alterations in alanine, aspartate, and glutamate metabolism (via citric acid and carbamoyl phosphate) and arginine biosynthesis indicate dynamic alignment of amino acid turnover with energy demand and nitrogen disposal [49 [↗](#), 50 [↗](#)]. RQ-associated enrichment of citrate cycle intermediates such as malate and citric acid further supports circadian gating of oxidative phosphorylation [51 [↗](#)]. Elevated levels of D-glucose were mapped to the KEGG pathway for neomycin, kanamycin, and gentamicin biosynthesis. While *Drosophila* does not synthesize these antibiotics, the enrichment likely reflects the presence of shared carbohydrate intermediates (e.g., glucose, glucose-6-phosphate, UDP-glucose) common to multiple biosynthetic processes. This mapping is more indicative of altered carbohydrate and amino sugar metabolism, or potentially reflects microbial contributions, rather than actual antibiotic production by the host [52 [↗](#), 53 [↗](#)]. Additional contributions from glyoxylate and dicarboxylate metabolism reinforce the role of anaplerotic flux and redox cycling [54 [↗](#)]. Notably, persistent correlations between RQ and pantothenate precursors suggest synchronized CoA biosynthesis, potentially coordinating fatty acid oxidation and TCA cycle input [55 [↗](#)]. Together, these results demonstrate that under LD conditions, wild-type flies sustain temporal coordination between respiration and mitochondrial metabolism, integrating redox homeostasis, nitrogen metabolism, and substrate availability in a circadian-dependent manner.

Altered Amino Acid Metabolism and Mitochondrial Dysregulation in *fmn* Mutants

The *fumin* (*fmn*) mutant, characterized by chronic sleep loss, exhibited genotype-specific disruptions in metabolic coordination, as revealed by metabolites whose temporal dynamics were strongly correlated with respiratory quotient (RQ). In *fmn* flies, significant RQ correlations ($p < 0.05$) were identified in pathways including arginine biosynthesis (citrulline, glutamine), purine metabolism (inosine, glutamine), and nitrogen metabolism-suggesting a decoupling of amino acid turnover and nucleotide cycling from respiratory demand [56 [↗](#), 57 [↗](#)]. Glutamine's strong association with RQ highlights its central role in buffering nitrogen flux and supporting biosynthetic processes under energetic stress [58 [↗](#)]. Correlated dynamics of 2-hydroxyglutarate, a key intermediate in butanoate metabolism, point to impaired TCA cycle flux and redox imbalance [59 [↗](#)]. Likewise, RQ-associated shifts in methylhistidine suggest altered histidine metabolism, potentially impacting mitochondrial protein turnover and methylation processes. Acylcarnitine metabolites, indicators of lipid oxidation, consistently emerged as strong correlates across all genotypes but particularly in *fmn*, underscoring lipid metabolism's critical role in respiratory coupling. Particularly in mutants, increased reliance on fatty acid oxidation may reflect compensatory responses to disrupted carbohydrate metabolism and sustained energetic demands. Together, these findings indicate a failure in aligning substrate oxidation with mitochondrial respiratory output in *fmn* mutants. Building on prior evidence of mitochondrial stress in sleep-

deprived *fnn* flies [60], our integrative respirometry-metabolomics analysis reveals disrupted coupling between energy metabolism and amino acid pathways, underscoring the physiological cost of chronic sleep loss on mitochondrial homeostasis.

Disrupted Carbon Metabolism and Mitochondrial Stress in *sss* Mutants

The *sleepless* (*sss*) mutant displayed significant disruptions in carbon and amino acid metabolism, as indicated by strong correlations between respiratory quotient (RQ) and metabolites involved in glycine, serine, and threonine metabolism (choline, dimethylglycine, glycine), butanoate metabolism (acetoacetate, succinate), and carbohydrate pathways [22]. Notably, RQ-associated shifts in choline and dimethylglycine suggest dysregulation of one-carbon metabolism and methyl group transfer, which can impair mitochondrial function and epigenetic stability [61, 62]. Correlated dynamics of acetoacetate, a ketone body often found in starvation, and succinate (both key intermediates of butanoate metabolism) point to altered TCA cycle input and mitochondrial redox imbalance [63, 64]. In parallel, strong RQ correlations with sucrose and D-glucose (from both starch/sucrose and galactose metabolism) indicate disrupted carbohydrate processing and inefficient substrate utilization [65]. cAMP and biotin were also found to associate with respiration, both unique in metabolic regulation through altered neuromodulatory signaling and cofactor metabolism. These findings suggest that *sss* mutants exhibit altered metabolic routing in response to respiratory demand, shifting substrate utilization toward carbohydrate and amino acid pathways under sleep-deprived conditions, consistent with metabolic stress responses observed in other models of sleep loss [66].

Loss of Temporal Coupling of Mitochondrial Metabolism in Clock Mutants

In *period* (*per*⁰¹) mutants lacking a functional circadian clock, the temporal coordination between metabolism and respiration was disrupted across nearly every measured metabolite through intermittent strong correlations with respiration, reflecting broad metabolic chaos [1, 67]. Strong correlations with RQ were observed in alanine, aspartate, and glutamate metabolism (L-alanine, glutamine, fumarate, oxoglutarate) and arginine biosynthesis, suggesting inefficient routing of amino acid-derived substrates into mitochondrial energy production [50]. Delayed RQ-associated dynamics of glutamine, oxoglutarate, and fumarate indicate a mismatch between substrate availability and respiratory output [2]. Additionally, metabolites from arginine and proline metabolism (creatine, spermidine, N-acetylputrescine, 4-hydroxyproline) exhibited altered RQ correlations, suggesting impaired mitochondrial redox buffering and compromised integrity [68]. Perturbations in purine metabolism-including adenosine, deoxyadenosine monophosphate, xanthosine, and ADP-ribose-further point to dysregulated ATP turnover and nucleotide imbalance. Correlations with quinolinic acid and niacinamide reflect altered nicotinate and nicotinamide metabolism, implicating disrupted NAD⁺ biosynthesis and redox state [48]. Additional RQ-associated changes in NADPH and TCA intermediates (fumarate, oxoglutarate) align with impaired glutathione metabolism and reduced mitochondrial capacity. Microbial-derived metabolites (trigonelline, phenylacetyl glycine) were also identified in the correlation analysis, suggesting potential gut microbiome interactions that become evident in circadian disruption. Together, these findings suggest that the loss of circadian timing in *per*⁰¹ mutants lead to widespread uncoupling of substrate utilization and mitochondrial respiration. This breakdown in temporal regulation disrupts energy homeostasis, highlighting the essential role of the circadian clock in coordinating metabolic flux with respiratory demand [1, 51, 69].

Metabolic Desynchrony and Redox Imbalance in Wild-Type Flies Under Constant Darkness

In wild-type flies maintained under constant darkness (DD), the absence of environmental light cues led to disrupted temporal alignment between mitochondrial respiration and metabolic pathways [70, 71]. Generally, we note a shift towards reactive metabolic adjustments (positive lags) compared to WT-LD instead of anticipatory preparation, indicating reduced metabolic efficiency. RQ-correlated metabolites showed significant enrichment in arginine biosynthesis (citrulline, ornithine, urea), glycine, serine, and threonine metabolism (L-cystathionine, phosphoserine, pyruvic acid), and cysteine and methionine metabolism-indicating altered integration of nitrogen and sulfur amino acid metabolism with respiratory activity [49]. Notably, the convergence of pyruvic acid across multiple pathways suggests dysregulated entry points into the TCA cycle, while the consistent association of L-cystathionine highlights impaired redox buffering and methylation capacity [72, 73]. Altered correlations in arginine and proline metabolism (ornithine, pyruvic acid) further underscore compromised coupling between amino acid turnover and mitochondrial energy production [29]. Altered lipid metabolism and increased nitrogen catabolism are likely compensatory mechanisms in the absence of environmental cues.

These findings build on previous reports of circadian desynchrony by providing direct functional evidence that environmental light cues are essential for maintaining coherent coordination between respiration and key metabolic circuits [50, 74]. The resulting metabolic misalignment under constant darkness emphasizes the circadian clock's dependence on external entrainment to regulate energy homeostasis at the organismal level [54].

Conclusions

This study demonstrates that sleep loss and circadian disruption impair metabolic homeostasis in *Drosophila melanogaster* via distinct yet converging mechanisms. Wild-type flies under light-dark cycles (WT-LD) maintained coordinated anticipatory respiratory and metabolic rhythms, while sleep mutants (*fmn*, *sss*) and circadian-disrupted flies (*per*⁰¹, WT-DD) showed altered substrate utilization, redox imbalance, and uncoupling of mitochondrial pathways, likely reactive to respiration. These findings highlight the essential role of both sleep and circadian timing in regulating energy metabolism. The presence of neuromodulatory metabolites (e.g., cAMP in *sleepless* mutants) and microbial-derived compounds (trigonelline, phenylacetylglycine in *per*⁰¹ mutants) points toward additional layers of metabolic regulation influenced by neuronal activity and gut microbiota interactions, revealing more complex metabolic dysregulation in sleep-deprived and circadian-disrupted states. By combining high-resolution respirometry with targeted metabolomics, we establish *Drosophila* as a powerful model for investigating the molecular basis of sleep- and clock-related metabolic dysfunction and potential therapeutic interventions.

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Additional information

Author Contributions

Conceptualization, F.A., and A.M.W.; methodology, F.A., D.M.M., P.H., J.K. and A.M.W.; formal analysis, F.A., A.S.G., and A.M.W.; investigation, F.A. and D.M.M.; resources, A.S. and A.M.W.; writing-original F.A.; writing-review and editing, F.A., A.S.G., J.K., A.S., and A.M.W.; visualization, F.A., A.S.G. and A.M.W.; supervision, A.M.W.; and funding acquisition, A.S. and A.M.W.

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Additional files

Supplementary figures [↗](#)

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Reviewer #1 (Public review):

Summary:

This study by Akhtar et al. aims to investigate the link between systemic metabolism and respiratory demands, and how sleep and the circadian clock regulate metabolic states and respiratory dynamics. The authors leverage genetic mutants that are defective in sleep and circadian behavior in combination with indirect respirometry and steady-state LC-MS-based metabolomics to address this question in the *Drosophila* model.

First, the authors performed respirometry (on groups of 25 flies) to measure oxygen consumption (VO₂) and carbon dioxide production (VCO₂) to calculate the respiratory quotient (RQ) across the 24-hour day (12h:12h light-dark cycle) and assess metabolic fuel utilization. They observed that among all the genotypes tested, wild type (WT) flies and *per0* flies in LD and WT flies in DD exhibit RQ >1. They concluded the >1 RQ is consistent with active lipogenesis. In contrast, the short-sleep mutants *fumin* (*fmn*) and *sleepless* (*sss*) showed significantly different RQ; the *fmn* exhibits a slight reduction in RQ values, suggesting increased reliance on carbohydrate metabolism, while *sss* exhibits even lower RQ (0.94), consistent with a shift toward lipid and protein catabolism.

The authors then proceeded to bin these measurements in 12-hour partitions, ZT0-12 and ZT12-24, to assess diurnal differences in average values of VO₂, VCO₂, and RQ. They observed significant day-night differences in metabolic rates in WT-LD flies, with higher rates during the day. The diurnal differences remain in the short-sleep mutants, but the overall metabolic rates are higher. WT-DD flies exhibit the lowest respiratory activity, although the day-night differences remain in free-running conditions. Finally, *per01* mutants exhibit no significant change in day-night respiratory rates, suggesting that a functional circadian clock is necessary for diurnal differences in metabolic rates.

They then performed finer-resolution 24-hour rhythmic analysis (RAIN and JTK) to determine if VO₂, VCO₂, and RQ exhibit 24-hour rhythmic and if there are genotype-specific differences. Based on their criteria, VCO₂ is rhythmic in all conditions tested, while VO₂ is rhythmic in all conditions except in *fmn*-LD. Finally, RQ is rhythmic in all 3 mutants but not in WT-LD and WT-DD. Peak phases for the rhythms were deduced using JTK lag values.

The authors proceeded to leverage a previously published steady-state metabolite dataset to investigate the potential association of RQ with metabolite profiles. Spearman correlation was performed to identify metabolites that exhibit coupling to respiratory output. Positive and negative lag analysis were subsequently performed to further characterize these associations based on the timing of the metabolite peak changes relative to RQ fluctuations. The authors suggest that a positive lag indicates that metabolite changes occur after shifts in RQ, and a negative lag signifies that metabolite changes precede RQ changes. To visualize metabolic pathways that exhibit these temporal relationships, a clustered heatmap and enrichment analysis were performed. Through these analyses, they concluded that both sleep and circadian systems are essential for aligning metabolic substrate selection with energy demands, and different metabolic pathways are misregulated in the different mutants with sleep and circadian defects.

Strength:

The research questions this study explores are significant, given that metabolism and respiratory demand are central to animal biology. The experimental methods used, including the well-characterized fly genetic mutants, the newly developed method for indirect calorimetry measurements, and LC-MS-based metabolomics, are all appropriate. This study provides insights into the impact of sleep and circadian rhythm disruption on metabolism and respiratory demand and serves as a foundation for future mechanistic investigations.

Weaknesses:

There are some conceptual flaws that the authors need to address regarding circadian biology, and some of the conclusions can be better supported by additional analysis to provide a stronger foundation for future functional investigation. At times, the methods, especially the statistical analysis, are not well articulated; they need to be better explained.

<https://doi.org/10.7554/eLife.108681.1.sa2>

Reviewer #2 (Public review):

This is an innovative and technically strong study that integrates dual-gas respirometry with LC-MS metabolomics to examine how sleep and circadian disruption shape metabolism in *Drosophila*. The combination of continuous O₂/CO₂ measurements with high-temporal-resolution metabolite profiling is novel and provides fresh insight into how wild-type flies maintain anticipatory fuel alignment, while mutants shift to reactive or misaligned metabolism. The use of lag-shift correlation analysis is particularly clever, as it highlights temporal coordination rather than static associations. Together, the findings advance our understanding of how circadian clocks and sleep contribute to metabolic efficiency and redox balance.

However, there are several areas where the manuscript could be strengthened. The authors should acknowledge that their findings may be gene-specific. Because sleep deprivation was not performed, it remains uncertain whether the observed metabolic shifts generalize to sleep loss broadly or are restricted to the *fmn* and *sss* mutants. This concern also connects to the finding of metabolic misalignment under constant darkness despite an intact clock. The conclusion that external entrainment is essential for maintaining energy homeostasis in flies may not translate to mammals. It would help to reference supporting data for the finding and discuss differences across species. Ideally, complementary circadian (light-dark cycle disruption) or sleep deprivation (for several hours) experiments, or citation of comparable studies, would strengthen the generality of the findings. Figures 1-4 are straightforward and clear, but when the manuscript transitions to the metabolite-respiration correlations, there is little description of the metabolomics methods or datasets, which should be clarified. The Discussion is at times repetitive and could be tightened, with the main message (i.e., wild-type flies align metabolism in advance, while mutants do not) kept front and center. Terms such as "anticipatory" and "reactive" should be defined early and used consistently throughout.

Overall, this is a strong and novel contribution. With clarification of scope, refinement of presentation, and a more focused Discussion, the paper will make a significant impact.

<https://doi.org/10.7554/eLife.108681.1.sa1>

Reviewer #3 (Public review):

Summary:

The authors investigate how sleep loss and circadian disruption affect whole-organism metabolism in *Drosophila melanogaster*. They used chamber-based flow-through respirometry to measure oxygen consumption and carbon dioxide production in wild-type flies and in mutants with impaired sleep or circadian function. These measurements were then integrated with a previously published metabolomics dataset to explore how respiratory dynamics align with metabolic pathways. The central claim is that wild-type flies display anticipatory coordination of metabolic processes with circadian time, while mutants exhibit reactive shifts in substrate use, redox imbalance, and signs of mitochondrial stress.

Strengths:

The study has several strengths. Continuous high-resolution respirometry in flies is challenging, and its application across multiple genotypes provides good comparative insight. The conceptual framework distinguishing anticipatory from reactive metabolic regulation is interesting. The translational framing helps place the work in a broader context of sleep, circadian biology, and metabolic health.

Weaknesses:

At the same time, the evidence supporting the conclusions is somewhat limited. The metabolomics data were not newly generated but repurposed from prior work, reducing novelty. The biological replication in the respirometry assays is low, with only a small number of chambers per genotype. Importantly, respiratory parameters in flies are strongly influenced by locomotor activity, yet no direct measurements of activity were included, making it difficult to separate intrinsic metabolic changes from behavioral differences in mutants. In addition, repeated claims of "mitochondrial stress" are not directly substantiated by assays of mitochondrial function. The study also excluded female flies entirely, despite well-documented sex differences in metabolism, which narrows the generality of the findings.

<https://doi.org/10.7554/eLife.108681.1.sa0>

Author response:

We thank the reviewers for their thoughtful public feedback. Our revision will clarify scope and methods/statistics, as well as streamline the narrative so the central message is clear: wild-type flies exhibit anticipatory alignment of fuel selection with circadian time, whereas short-sleep and clock mutants show reactive or misaligned metabolism under our conditions.

Major conceptual and experimental revisions:

(1) We will define “anticipatory” (clock-aligned, pre-emptive substrate choice) and “reactive” (post-hoc substrate shifts) up front and use these terms consistently. We will clearly distinguish diurnal (LD) from circadian (DD) regulation and avoid implying that DD abolishes rhythmicity. Claims will be limited to the tested genotypes (*fmn*, *sss*, and *per⁰¹*) without generalizing to all forms of sleep loss or to mammals (although we will speculate in the discussion about translation and generalizability). We will temper language around external entrainment in DD to “contributes strongly under our conditions in flies.”

(2) We will expand the respirometry and rhythmicity sections (RAIN/JTK parameters, period/phase outputs, multiple-testing control). We will clarify that each measurement is an average of 300 flies per genotype (25 flies/chamber, 4 chambers/experiment, 3 experimental days) and specify the chamber as the experimental unit with *n* and error structure in each figure legend. For metabolomics–respirometry correlations, we will briefly describe dataset parameters, time-matching across ZT, normalization, Spearman correlations, and lag interpretation.

(3) We are performing additional experimental measurements through tissue respirometry of gut tissues and ROS staining to support our claims of “mitochondrial stress” in the short sleeping mutants. We note that this has already been shown for *fmn* in Vaccaro et al (Cell, 2020) and we will extend this to the other mutants studied in our work.

Reviewer-specific points

Reviewer #1.

We will clarify the circadian/diurnal framing, fully report rhythmicity analyses (parameters, n, q-values, phases), and better explain the metabolomics-respiration coupling with a concise workflow figure and supplementary table. The conclusion that sleep and clock systems align substrate selection with energy demand will be presented as supported under our tested conditions and positioned as groundwork for future mechanistic studies.

Reviewer #2.

We will state explicitly that findings may be gene-specific and avoid inferring generality to all sleep loss. We will soften cross-species language about external entrainment and add a brief note on species differences. For behavioral context (activity/feeding/sleep in *fmn* and *sss*), we will cite our related manuscript in revision (Malik et al, <https://www.biorxiv.org/content/10.1101/2023.10.30.564837v2>) in which we have measured both activity and feeding for *fmn*, *sss*, and wt flies. We will add a concise description of LC-MS processing and pathway analysis and define “anticipatory”/“reactive” early, using them consistently.

Reviewer #3.

We acknowledge that metabolomics were repurposed and emphasize the novelty of integrating continuous VCO₂ and VO₂ respirometry with temporal lag analysis. We will report replication clearly (chambers as the unit, n per genotype) and acknowledge locomotor activity as a potential confound, pointing to the related manuscript (Malik et al) for independent activity/feeding measurements and experimental measures of mitochondrial stress as outlined above. We will also further note that only males were studied, outlining this as a limitation and a future direction.

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